

ST-911 — Completion Summary

Ticket: ST-911 — Telemetry V5 Schema Sync
Version: v1.0.1-beta
Author: Mike (Lead Developer / Build Agent)
Date: 2026-05-14
Status:  Complete — Awaiting Orchestrator ACK for merge

Deliverables

#	Deliverable	File	Status
1	Schema Delta Analysis	ST-911_SCHEMA_DELTA_ANALYSIS.md	 Complete
2	Engine Parser Update (V5 COL_MAP)	nextjs_space/lib/s2s-engine/telemetry-parser.ts	 Complete
3	Legacy Parser Update (V5 COL_MAP)	nextjs_space/lib/telemetry-parser.ts	 Complete
4	Telemetry Types Update	nextjs_space/types/telemetry.ts	 Complete
5	Integration Smoke Test	tests/integration/smoke_v5_csv_tests	 Complete
6	Path View Impact Statement	ST-911_PATH_VIEW_IMPACT.md	 Complete
7	Changelog Entry (v1.0.1-beta)	PRO-JECT_CHANGELOG.md	 Complete

Schema Changes Made

Critical Column Renames (V4 → V5)

V4 Header	V5 Header	Internal Key
alt_ft	alt_msl_ft	alt
agl_ft	alt_agl_ft	agl
heading_deg	hdg_mag_deg	hdg

New V5 Fields Added

Seven new columns from the V5 recorder schema were added to the COL_MAP with appropriate internal key mappings, ensuring forward compatibility with enhanced telemetry data.

Backward Compatibility

Both V4 and V5 column headers are accepted. The parser's COL_MAP includes aliases for renamed columns, so older CSV files continue to parse without error.

Test Results

Smoke Test: `evaluateFlight()`

- **Result:**  PASS (25/25 assertions)
- **Test vectors:**
 - `s2s_telemetry_20260511_173825.csv`
 - `s2s_telemetry_20260511_204358.csv`
- **Coverage:** Full parse → phase detection → grading pipeline round-trip

Validations Confirmed

- V5 CSVs parse without “Missing Column” errors
 - All internal keys resolve correctly after rename mapping
 - Downstream graders (`phase-detector` , `arc1-grader` , `landing-scorer`) consume data without changes
 - No regressions in V4 CSV parsing
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Files Modified

File	Change Type	Description
<code>nextjs_space/lib/s2s-engine/telemetry-parser.ts</code>	Modified	Updated COL_MAP with V5 column renames and new fields
<code>nextjs_space/lib/telemetry-parser.ts</code>	Modified	Updated legacy parser COL_MAP for V5 compatibility
<code>nextjs_space/types/telemetry.ts</code>	Modified	Added new V5 field definitions to <code>TelemetryRow</code> interface
<code>tests/integration/smoke_v5_csv_test.ts</code>	Created	New smoke test for V5 CSV end-to-end validation
<code>ST-911_SCHEMA_DELTA_ANALYSIS.md</code>	Created	Schema delta analysis and migration guide
<code>ST-911_PATH_VIEW_IMPACT.md</code>	Created	Path View Impact statement (no gating changes)
<code>ST-911_COMPLETION_SUMMARY.md</code>	Created	This document
<code>PROJECT_CHANGELOG.md</code>	Modified	Added v1.0.1-beta entry for ST-911

Risk Assessment

- **Risk Level:** High (telemetry/scoring pipeline)
- **Mitigation:** Backward-compatible aliases ensure V4 CSVs continue to work. Smoke test validates full scoring pipeline round-trip.
- **Revert Plan:** Restore prior COL_MAP entries in both `telemetry-parser.ts` files and remove new `telemetry.ts` fields.

Path View Impact

No changes to scoring logic, lesson gating, or student progression. All downstream graders consume via internal keys and require no logic changes. Lesson gating rules (score ≥ 3 , quiz $\geq 80\%$, debrief viewed) remain unchanged.